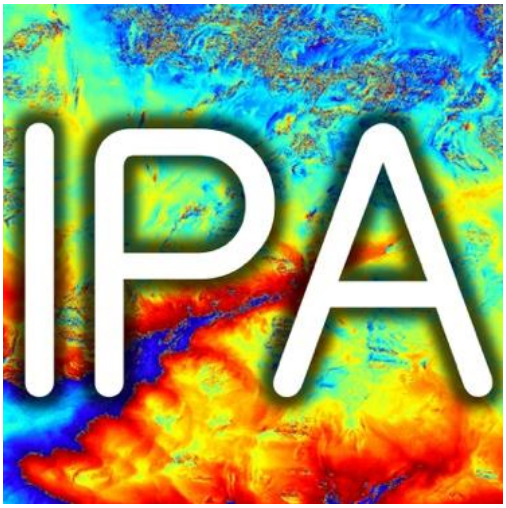


IPA

InSAR Product Analysis

QGIS Plugin – User Manual

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Authors: Fabio Bovenga - *National Research Council of Italy, IREA Institute, Bari (Italy)*
fabio.bovenga@cnr.it

Fabio Piccolino - *Polytechnic University of Bari – DICATECh, Bari (Italy)*
fabiopiccolino02@gmail.com

Table of Contents

Table of Contents 1
1. Introduction 2
1.1 Contributions and Acknowledgments 2
1.2 References 3
1.3 License 3
2. System Requirements 4
3. Installation 5
Method 1: Download from the QGIS Plugin Repository (*coming soon*) 5
Method 2: Install the development version from a ZIP file 5
4. Plugin Interface Overview 6
4.1 Mandatory Satellite Orbit Setup 6
4.2 Built-in Satellite Mission Presets 6
4.3 Background Processing and Responsiveness 7
4.4 Smart Layer Selection and Automatic Styling 7
5. IPA Modules 8
5.1 InSAR Feasibility Analysis 8
5.1.1 Sat. Orbit 8
5.1.2 Geo. Distortion Index (GDI) 9
5.1.3 Layover/Shadow Mask 10
5.1.4 Slope2LOS 11
5.2 InSAR Product Analysis 12
5.2.1 Sat. Orbit 12
5.2.2 InSAR Sensitivity Map (ISM) 12
5.3 MTInSAR Product Analysis 13
5.3.1 Sat. Orbit 13
5.3.2 Geo. Distortion Index (GDI) 14
5.3.3 Slope Vel. 15
5.3.4 Spatial Analysis 17
5.3.5 Temporal Analysis 18
Visualization 20

1. Introduction

IPA – InSAR Product Analysis is a QGIS plugin designed to support the analysis of products derived through SAR Interferometry (InSAR) techniques. Specifically, the plugin enables preliminary assessments of the suitability of a given area for InSAR analysis and assists in the interpretation of InSAR and Multi-Temporal InSAR (MTInSAR) products through a set of integrated operational modules.

The plugin is built entirely in Python on top of the PyQGIS API and combines NumPy- and SciPy-based scientific computing with a native PyQt/QGIS graphical interface. Computationally intensive routines — statistical regressions, raster scans, ray-tracing algorithms — run in dedicated background threads, so the QGIS map canvas and legend remain fully responsive even while processing large raster grids or dense MTInSAR point catalogues (see Section 4 for details on the interface behavior).

The plugin is organized into three main modules, all accessible from a single unified window through a tabbed interface:

- **Module 1: InSAR Feasibility Analysis** – preliminary DEM-based assessment of the geometric suitability of an area for InSAR monitoring, usable even before any InSAR product is available.
- **Module 2: InSAR Product Analysis** – post-processing evaluation of the reliability of individual differential interferograms by combining geometric and interferometric information.
- **Module 3: MTInSAR Product Analysis** – post-processing analysis of MTInSAR products (coherent target locations, LOS displacement time series and their mean rate values), including spatial distribution analysis and time-series characterization of displacement trends.

The goal is to provide a flexible and integrated environment that enhances the interpretation of interferometric results, supports decision-making, and facilitates the extraction of meaningful information from satellite-based ground deformation measurements.

IPA is intended for use by geologists, geotechnical engineers, remote sensing specialists, and researchers working with InSAR data for applications such as landslide monitoring, subsidence analysis, infrastructure stability assessment, and environmental monitoring.

The plugin is developed as open-source software and is intended for publication in the official QGIS Plugin Repository, making it freely available to the wider community of GIS and remote sensing practitioners.

1.1 Contributions and Acknowledgments

Fabio Bovenga conceived the plugin and developed its specific functionalities along with their mathematical formulations. Fabio Piccolino implemented these functionalities in Python. Together, they tested the plugin using various SAR data and ground configurations, and they are responsible for its ongoing updates and maintenance.

The development of IPA QGIS plugin was supported in part by Regione Puglia (Italy) under project “Utilizzo di intelligenza artificiale e dati satellitari per il monitoraggio dell’instabilità del territorio”, POC PUGLIA FESR-FSE 2014 / 2020 – Programma Regionale RIPARTI, grant agreement 01975b92; and in part by the European Union – Next Generation EU, Mission 4, Component 2, CUP

H53D23001660006 (PRIN22 Project "MIRAGE: Mass movement Investigation and prediction through geomorphology, Remote sensing and Artificial intelligence").

1.2 References

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Bovenga F., Piccolino, F., 2025. InSAR Product Analysis (IPA): a QGIS tool for slope instability assessment based on SAR interferometry. EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-17224, https://doi.org/10.5194/egusphere-egu25-17224, 2025.

Bovenga F., Argentiero I., Piccolino F., 2026°. Methodologies and functionalities for a QGIS-based analysis of DInSAR/MTInSAR products in 13th International Workshop on "Advances in the Science and Applications of SAR Interferometry." – FRINGE 2026, Jagiellonian University, Krakov, Poland, 15-19/06/2026.

Bovenga F., Piccolino, F., Argentiero I., 2026b. A QGIS plugin module for performing InSAR-based slope ground instability assessment. In preparation for submission to GIScience & Remote Sensing.

Dixon P. M., 2002. Ripley's K-function. In Encyclopedia of Environmetrics, 1796–1803. Wiley 2024, 1-7. DOI: 10.1109/IHTC61819.2024.10855063.

Mondini, A. C., Simoni, A., Bovenga, F., Mercurio, A., Reyes-Carmona, C., Boyun, Y., Agliardi, F., 2026. Automated landslide detection in SAR wrapped interferograms using a geomorphology-constrained YOLO CNN. Preprint at https://doi.org/10.31223/X51B7T.

1.3 License

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2. System Requirements

The following table lists the plugin's minimum system requirements:

Component	Minimum Version
QGIS	3.34.4 – Prizren
Python	3.9.5
GDAL library	3.3
NumPy library	1.20
SciPy library	1.6

NumPy and SciPy are used, respectively, for the vectorized array computations that underpin all raster and vector algorithms and for the numerical routines (e.g. non-linear curve fitting) used in the Temporal Analysis functionality.

3. Installation

Method 1: Download from the QGIS Plugin Repository

- Open QGIS.
- Go to *Plugins* > *Manage and Install Plugins…*
- In the *All* tab of the Plugin Manager, type *IPA – InSAR Product Analysis* in the search bar.
- Select the *IPA – InSAR Product Analysis* plugin from the list and click *Install Plugin*.

Method 2: Install the development version from a ZIP file

- Go to the GitHub repository: https://github.com/PluginIPA/IPA
- Download the plugin repository as a ZIP archive.
- Open QGIS.
- Go to *Plugins* > *Manage and Install Plugins…*
- Click on the *Install from ZIP* tab.
- Select the downloaded ZIP file and click *Install Plugin*.

4. Plugin Interface Overview

IPA centralizes all its functionalities in a single main window rather than opening a separate dialog for each operation. Access to the different functions is governed by a tabbed interface: a first row of tabs switches between the three main modules (InSAR Feasibility Analysis, InSAR Product Analysis, MTInSAR Product Analysis), and a second, module-specific row of tabs switches between the individual functionalities of the selected module.

When the plugin is opened, the interface is preset on the first module, InSAR Feasibility Analysis, on its Sat. Orbit tab.

4.1 Mandatory Satellite Orbit Setup

Each of the three modules includes its own Satellite Orbit Setup (Sat. Orbit) tab, and its configuration is a blocking prerequisite: until the satellite acquisition geometry has been defined and validated in that tab, all remaining tabs and "Run" buttons of the corresponding module stay disabled, and any attempt to execute a function returns an on-screen error.

To configure the orbit, the user must:

- select the orbit pass direction (***ASC*** = ascending, ***DESC*** = descending);
- either select a reference satellite mission from the built-in preset list (Table 1) or manually enter the ***Sat. Height***, ***Incidence Angle***, and ***Orbits per Day*** parameters.

Once the parameters are entered, click Upload. If the upload is successful, the Uploaded checkbox is automatically checked and the remaining tabs of that module become active.

4.2 Built-in Satellite Mission Presets

The plugin ships with a built-in dictionary of nominal orbital parameters for the main current and past SAR missions, available from the Sat. Orbit tab of every module. Selecting a mission automatically fills in the orbit height, incidence angle, and orbits-per-day fields; the wavelength value is used internally by some functions (e.g. Temporal Analysis).

Satellite Mission	Orbit Height (km)	Incidence Angle (deg)	Orbits per Day	Wavelength (mm)
Sentinel-1	693.000	98.18	175 / 12	56
Cosmo-SkyMed / CSG	619.600	97.86	237 / 16	31
TerraSAR-X / TanDEM-X	514.000	97.44	167 / 11	31
ALOS PALSAR (ALOS-1)	691.650	98.16	671 / 46	236.2
ALOS-2 PALSAR-2	628.000	97.90	207 / 13	236
SAOCOM	619.600	97.89	237 / 16	235.1
ENVISAT	799.800	98.55	501 / 36	56.3
ERS 1/2	780.000	98.50	501 / 36	56.6

Table 1 – Satellite mission presets included in the plugin

4.3 Background Processing and Responsiveness

Heavy computations such as statistical regressions, radial ray-tracing over raster grids, and large vector attribute update are executed asynchronously in dedicated background threads, so the QGISMap Canvas and Legend remain fully responsive while a function is running. A progress bar and status label report the current progress and a stop control lets the user safely interrupt a running computation at any time, without leaving the project or the underlying data in an inconsistent state.

4.4 Smart Layer Selection and Automatic Styling

The layer-selection combo boxes used throughout the interface are automatically filtered to display only layers of the compatible data type: raster layers in the predictive (DEM-based) functions and vector layers in the post-processing functions operating on MTInSAR datasets. Where relevant, numeric parameters are pre-filled with nominal default values derived from the scientific literature, which the user is free to adjust to the specific characteristics of the study site.

Every output layer produced by the plugin is automatically added to the QGIS project and styled with a default, pre-configured symbology (grayscale ramp, color ramp, or categorical classification) appropriate to the parameter being mapped, so results can be interpreted immediately without any manual styling step.

5. IPA Modules

The plugin is organized into three main modules, each containing several functionalities, described in detail in the following sections.

5.1 InSAR Feasibility Analysis

This module allows a preliminary assessment of the potential quality of subsequent InSAR products, based on satellite orbital configurations and ground morphological characteristics derived from the Digital Elevation Model (DEM). It is organized in four sub-functions accessible through dedicated tabs: Sat. Orbit, Geometric (Geo.) Distortion Index (GDI), Layover/Shadow Mask, and Slope2LOS.

5.1.1 Sat. Orbit

This functionality allows users to select the satellite orbit mode (***ASC*** = ascending, ***DESC*** = descending) and the reference satellite mission. Alternatively, if the mission is not listed, users can manually enter the orbital parameters.

- ***Sat. Height***: satellite flight altitude.
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees.
- ***Orbits per Day***: satellite revisit time, expressed as a decimal number or an orbit/day ratio (e.g., 175/12).

Once the parameters are entered, click the Upload button. If the upload is successful, the Uploaded checkbox will be automatically checked (see Section 4.1).

Output:

This function does not produce a direct output file; it loads and verifies the input orbital parameters to be used in the other sub-functions of this module.

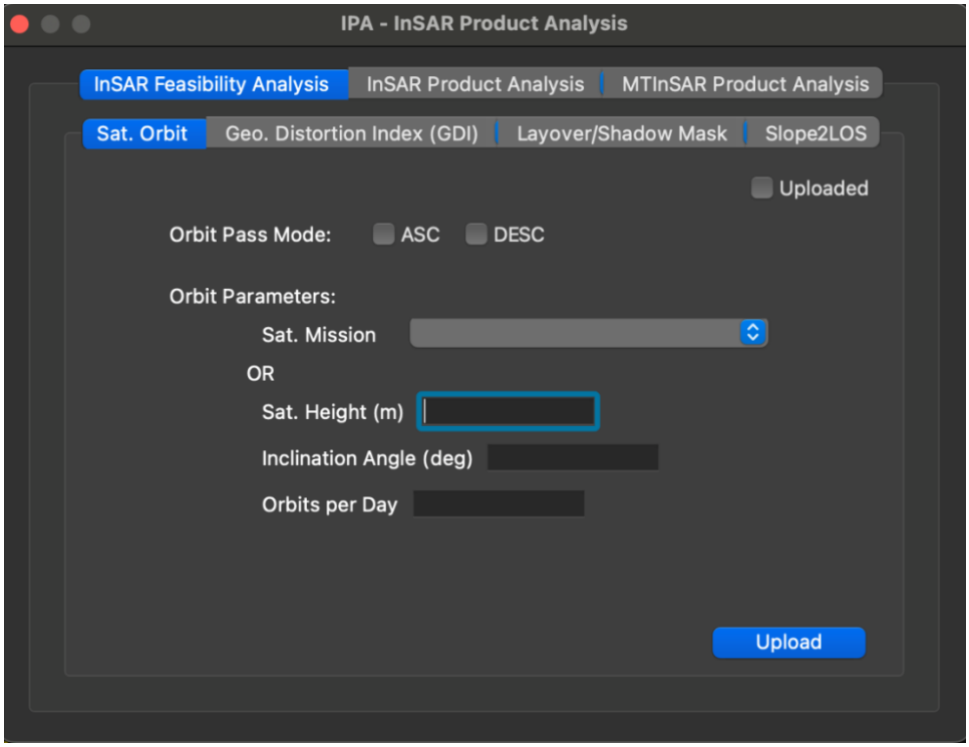


Figure 1 *– Main plugin window (hub) on startup, Sat. Orbit tab of Module 1*

5.1.2 Geo. Distortion Index (GDI)

This functionality evaluates the geometric distortion introduced by the terrain relief on the SAR image. It computes the GDI value based on the local slope and aspect of the DEM, as well as the satellite Line Of Sight (LOS) (Bovenga et al., 2026a; 2026b). The resulting GDI ranges between -1 and 1, with low distortions (good visibility) occurring for values close to 1 and strong distortions leading to unreliable SAR imaging occurring for values lower than 0.

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a GeoTIFF raster layer or entered manually as a fixed value;
- ***AOI Polygon Layer***: an optional polygon layer to perform the analysis within a specific area (e.g., an active landslide body or a rock glacier).

If a GeoTIFF incidence angle layer is provided, the plugin automatically checks that it is spatially consistent with the DEM (same extent, cell resolution, and coordinate reference system) before proceeding.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function outputs a single-band, continuous grayscale raster layer whose extent matches the selected polygon (if provided), the incidence angle GeoTIFF (if available), or otherwise the DEM. Each pixel in the output raster contains the GDI value, representing the local geometric sensitivity of that location to InSAR analysis.

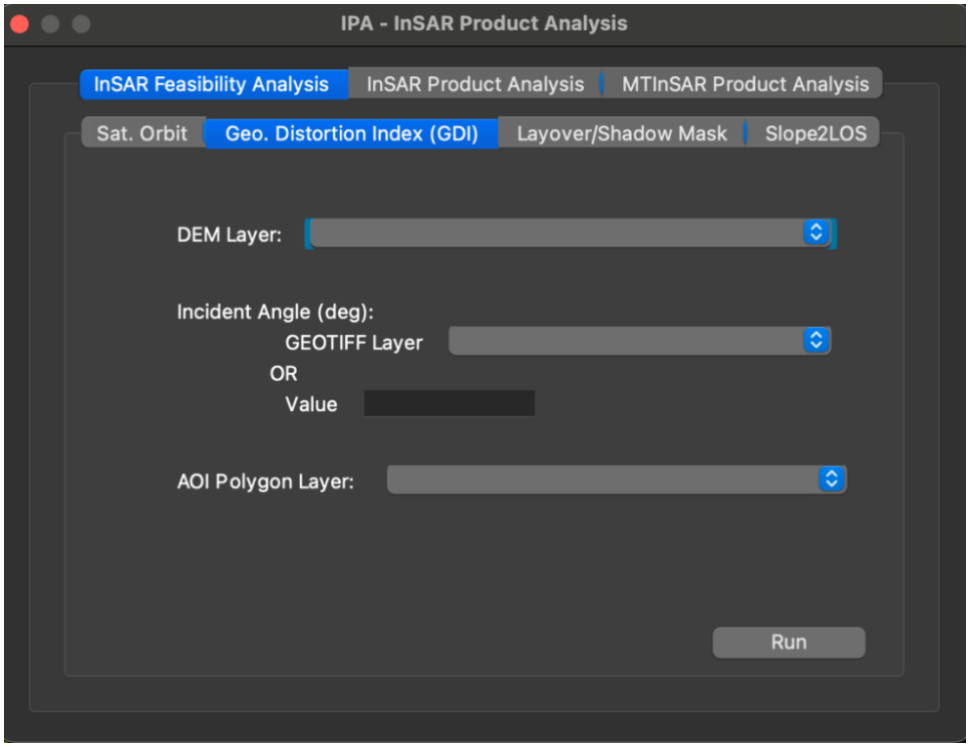


Figure 2 *– InSAR Feasibility Analysis window – GDI tab*

5.1.3 Layover/Shadow Mask

This functionality identifies the portions of terrain that are geometrically unusable for InSAR analysis because of the unfavorable acquisition geometry: shadow areas, where the radar signal cannot reach the ground, and layover areas, where the perspective of the wavefront is inverted by steep, radar-facing slopes.

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a GeoTIFF raster layer or entered manually as a fixed value;
- ***AOI Polygon Layer***: an optional polygon layer to perform the analysis within a specific area (e.g., an active landslide body or a rock glacier);
- orbital parameters (plane inclination and pass direction), set in the Sat. Orbit tab.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function returns a categorical raster mask with three classes: favorable visibility, shadow, and layover. The scanning direction of the underlying ray-tracing algorithm automatically adapts to the selected orbit mode (left-to-right for ascending passes and right-to-left for descending passes) so that the resulting mask correctly reflects the actual illumination geometry of the radar antenna. A default categorical style is applied to the output.

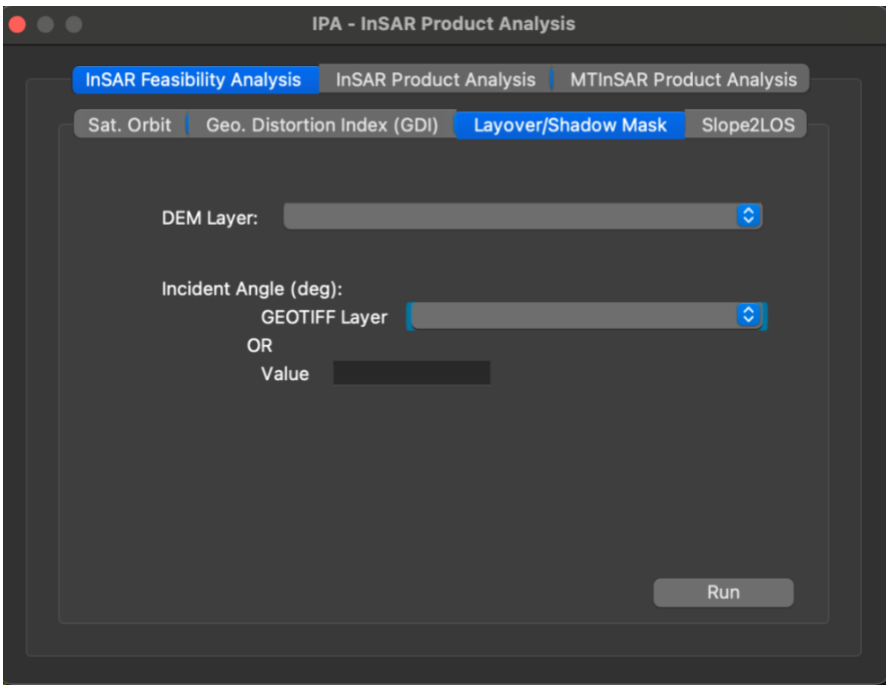


Figure 3 *– InSAR Feasibility Analysis window – Layover/Shadow Mask tab*



Figure 4 *– Default visibility classes applied to a Layover/Shadow Mask output*

5.1.4 Slope2LOS

This functionality computes the Slope2LOS parameter, a dimensionless coefficient representing the projection of the local terrain slope unit vector onto the SAR LOS.

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a GeoTIFF raster layer or entered manually as a fixed value;
- ***AOI Polygon Layer***: an optional polygon layer to perform the analysis within a specific area (e.g., an active landslide body or a rock glacier);
- ***Slope Threshold (deg)***: an optional slope threshold value, expressed in degrees, below which pixels are excluded from the computation.

The slope threshold allows excluding areas on flat or gently sloping terrain, where the assumption that displacement occurs along the direction of maximum slope loses physical meaning, and the Slope2LOS factor approaches zero, which would otherwise introduce unstable or meaningless results.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function outputs a continuous raster layer whose extent matches the selected polygon (if provided), the incidence angle GeoTIFF (if available), or otherwise the DEM. Each pixel contains the computed Slope2LOS value, ranging between -1 and 1.

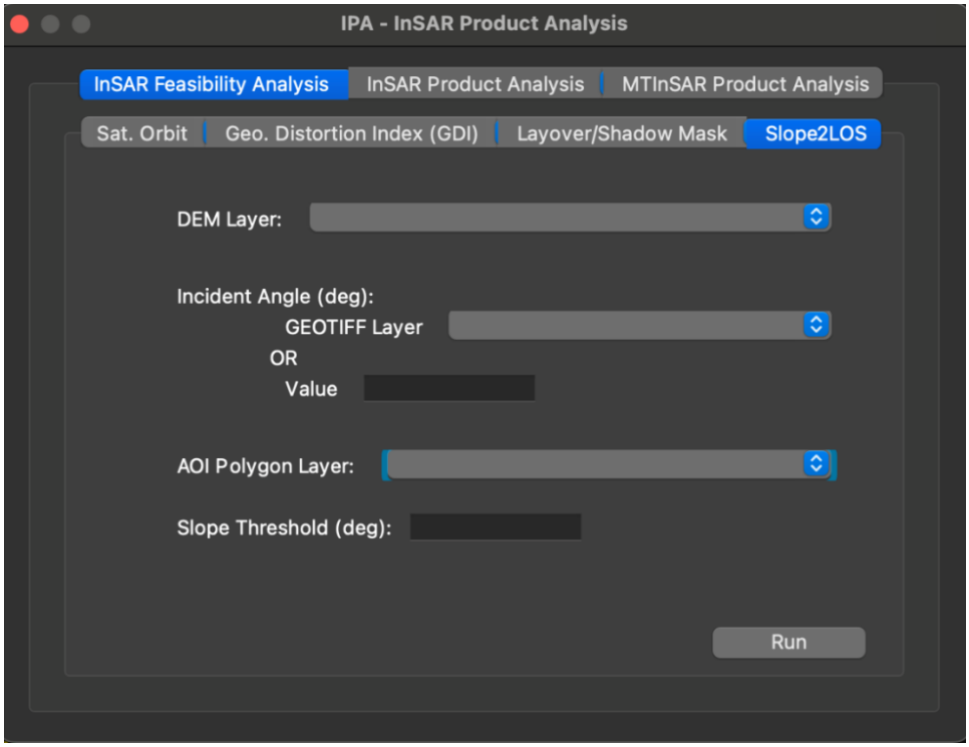


Figure 5 *– InSAR Feasibility Analysis window – Slope2LOS tab*

5.2 InSAR Product Analysis

This module allows the evaluation of the reliability of an available interferogram by combining geometric and interferometric information. It is designed for post-processing analyses based on already available InSAR products, and it is organized in two tabs: Sat. Orbit and InSAR Sensitivity Map (ISM).

5.2.1 Sat. Orbit

This functionality allows users to select the satellite orbit mode (***ASC***=ascending, ***DESC***=descending) and the reference satellite mission. Alternatively, if the mission is not listed, users can manually enter the orbital parameters.

- ***Sat. Height***: satellite flight altitude.
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees.
- ***Orbits per Day***: satellite revisit time, expressed as a decimal number or an orbit/day ratio (e.g., 175/12).

Once the parameters are entered, click the Upload button. If the upload is successful, the Uploaded checkbox will be automatically checked.

Output:

This function does not produce a direct output file; it loads and verifies the input orbital parameters to be used by the ISM sub-function.

5.2.2 InSAR Sensitivity Map (ISM)

This functionality produces an InSAR Sensitivity Map (ISM) by combining the Slope2LOS geometric factor with the local InSAR coherence, optionally weighted through a sigmoid function (Bovenga et al., 2025; Bovenga et al., 2026a; Mondini et al., 2026). The resulting map provides a spatial estimate of the reliability of the phase measurements, supporting the interpretation of deformation signals and the identification of unstable areas. Specifically, ISM values range from 0 and 1, representing weights of the corresponding InSAR phase values, where 0 means a useless phase (due to noise and/or geometrical distortions), while 1 means noise-free and ideal viewing conditions.

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***Coherence GeoTIFF Layer***: InSAR coherence map (derived from a standard InSAR processing) selectable from a GeoTIFF raster layer;
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a GeoTIFF raster layer or entered manually as a fixed value;
- ***Weighting Function Parameters***: optional parameters defining a function that weights the InSAR coherence values:
    - if "Unweighted" is selected, no coherence weighting is applied;
    - if "Weighted" is selected, a sigmoid function is used to weight the coherence values, suppressing the residual noise from low-coherence areas (e.g. vegetated land) and emphasizing coherent targets such as rock outcrops, urban areas, and infrastructures.

When weighting is enabled, two parameters control the shape of the sigmoid function:

- ***Steepness***: value of the steepness parameter, which governs the slope and the responsiveness of the sigmoid curve;
- ***Midpoint***: value of the sigmoid midpoint, which defines the inflection point and the critical coherence threshold.

If these text fields are left empty, default parameters are used: steepness = 15, midpoint = 0.4. If the study site requires a different calibration, the operator can manually override these values.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

This function produces a single-band, continuous raster layer normalized between 0 and 1. Each pixel contains the InSAR sensitivity value computed at that location, reflecting the combined effect of geometric sensitivity (Slope2LOS) and InSAR coherence (optionally weighted).

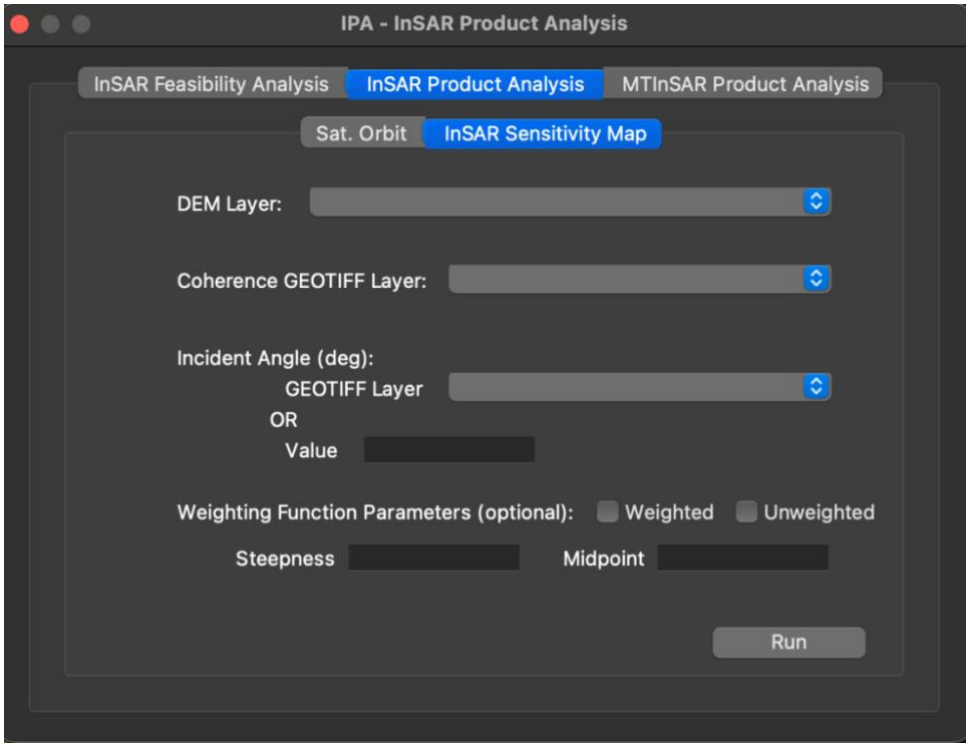


Figure 6 *– InSAR Product Analysis window – ISM tab*

5.3 MTInSAR Product Analysis

This module allows performing spatial and temporal analysis of existing MTInSAR products (e.g. coherent target locations, LOS displacement time series and their mean rate values) by combining geometric and interferometric information. The module operates directly on point vector layers is organized into five tabs: Sat. Orbit, GDI, Slope Vel., Spatial Analysis and Temporal Analysis.

5.3.1 Sat. Orbit

This functionality allows users to select the satellite orbit mode (***ASC*** = ascending, ***DESC*** = descending) and the reference satellite mission. Alternatively, if the mission is not listed, users can manually enter the orbital parameters.

- ***Sat. Height***: satellite flight altitude.
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees.

- ***Orbits per Day***: satellite revisit time, expressed as a decimal number or an orbit/day ratio (e.g., 175/12).

Once the parameters are entered, click the Upload button. If the upload is successful, the Uploaded checkbox will be automatically checked.

Output:

This function does not produce a direct output file; it loads and verifies the input orbital parameters to be used by the remaining sub-functions of this module.

5.3.2 Geo. Distortion Index (GDI)

This functionality computes the GDI value (introduced in Section 5.1.2) for each coherent target in the MTInSAR dataset, allowing filtering out those located in geometrically unfavorable positions.

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***MTInSAR Product Layer*:** vector layer containing the MTInSAR products (e.g. coherent target locations, mean rate LOS displacement values in millimetres per year, LOS displacement time series expressed in millimetres);
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a field in the MTInSAR layer or entered manually as a fixed value.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function returns a vector layer, a copy of the input MTInSAR layer that also includes the following attribute fields:

- SLOPE: local slope value derived from the DEM layer;
- ASPECT: local terrain aspect derived from the DEM layer;
- temp_los; heading angle calculated from the orbital inclination, pass mode (ascending or descending), and local geographic coordinates;
- GDI: GDI value.

A default symbology is automatically applied to classify the points based on their GDI values.

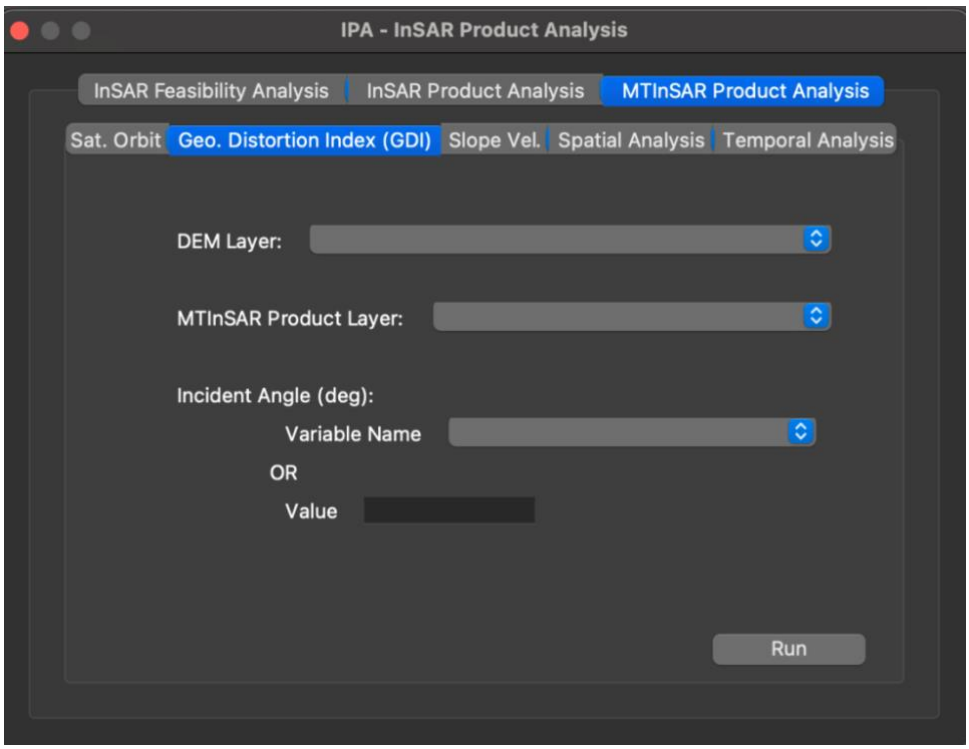


Figure 7 *– MTInSAR Product Analysis window – GDI tab*



Figure 8 *– Default style applied to a Module 3 GDI output*

5.3.3 Slope Vel.

This functionality converts the mean LOS displacement rate (or velocity) values into downslope mean displacement rate, based on the Slope2LOS factor, which depends on local terrain morphology and the satellite acquisition geometry. The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***MTInSAR Product Layer*:** vector layer containing the MTInSAR products (e.g. coherent target locations, mean rate LOS displacement values in millimetres per year, LOS displacement time series in millimetres);
- ***LOS Vel. Name***: name of the field corresponding to the LOS velocity expressed in millimetres per year (mm/y) and selected from the list of MTInSAR layer fields;
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a field in the MTInSAR layer or entered manually as a fixed value;
- ***Slope Threshold (deg)***: an optional slope threshold value, expressed in degrees, used to exclude coherent targets located on terrain with slope values below the threshold to avoid mathematically unstable projections.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function returns a vector layer named VSlope and a copy of the input MTInSAR dataset, which also includes the following attribute fields:

- SLOPE: local slope value derived from the DEM layer;
- ASPECT: local terrain aspect derived from the DEM layer;
- temp_los; heading angel calculated from the orbital inclination, pass mode (ascending or descending), and local geographic coordinates;
- L2A: LOS-to-Aspect angle. It represents the relative horizontal angle between the radar ground look direction (γ) and the terrain aspect (δ), used directly in the calculation of the Slope2LOS factor;
- SLOPE2LOS: dimensionless projection factor (ranging from -1 to 1) representing the scalar product between the downslope unit vector and the radar LOS unit vector, used to convert LOS displacement into downslope displacement;
- V_SLOPE: downslope mean displacement rate value expressed in millimetres per year (mm/y).

The computation is performed in a single write cycle directly on the layer's data provider, minimizing processing time. A default symbology is applied to the output layer to visually classify the coherent targets based on their V_SLOPE values, facilitating the interpretation of slope-parallel displacement trends.

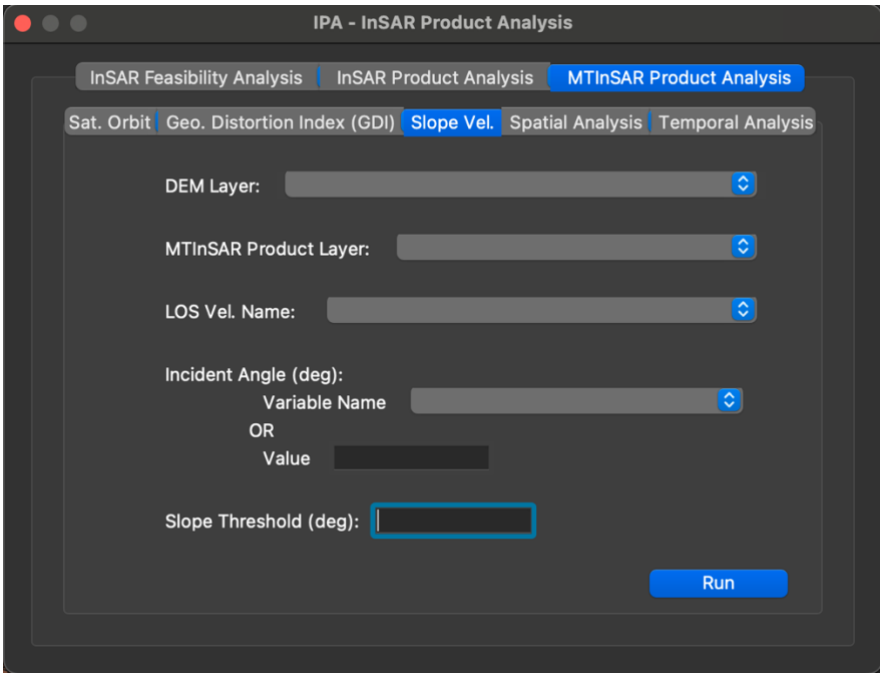


Figure 9 *– MTInSAR Product Analysis window – Slope Vel. Tab*



Figure 10 *– Default style applied to a Slope Vel. output*

5.3.4 Spatial Analysis

This functionality characterizes the spatial distribution of coherent targets within user-defined polygons, combining a point-density index with the Clark–Evans Nearest-Neighbor Index (Bovenga et al., 2026a; 2026b; Dixon, 2002) to assess whether the coherent target spatial distribution within the area of interest is uniform or includes empty regions (or voids).

The following inputs are required:

- ***DEM Layer***: digital elevation model in meters from a raster layer;
- ***MTInSAR Product Layer*:** vector layer containing the MTInSAR products (e.g. coherent target locations, mean rate LOS displacement values in millimetres per year, LOS displacement time series in millimetres);
- ***Incidence Angle (deg)***: incidence angle expressed in decimal degrees, selectable from a field in the MTInSAR layer or entered manually as a fixed value;
- ***AOI Polygon Layer***: an optional polygon layer to perform the analysis within a specific area (e.g., an active landslide body or a rock glacier);
- ***Product Spacing (m)***:
    - ***Range***: range spacing (in meters) of the MTInSAR products;
    - ***Azimuth***: azimuth spacing (in meters) of the MTInSAR products.

Polygons that are not entirely contained within the spatial extent of the DEM are automatically excluded from the computation to guarantee the consistency of the extracted indices.

Note: Before using this function, it is necessary to complete the Sat. Orbit tab of this module.

Output:

The function returns the input polygon layer, renamed a Spatial_Density, which also includes the following attribute fields:

- Density: spatial density of coherent targets within the polygon;
- PCT: percentage of the area of interest (single polygon), surface covered by coherent targets CTs (Bovenga et al., 2026a; 2026b);
- R: Clark–Evans Nearest-Neighbor Index indicating whether the coherent targets are clustered ($R \to 0$), randomly distributed ($R \to 1$), or uniformly distributed ($R \to 2.149$) within the polygon (Dixon, 2002);
- Z: Z-score expressing the statistical significance of the Clark-Evans coefficient with $Z < -1.92$ significance of clustering distribution and $|Z| < 1.92$ means significance of random distribution (Dixon, 2002);
- Void: binary flag (1/0) that automatically identifies the presence of data voids within the polygon perimeter by testing the following rule: $R < 0.8$.

A default color-ramp style, proportional to the PCT value, is applied to the output layer.

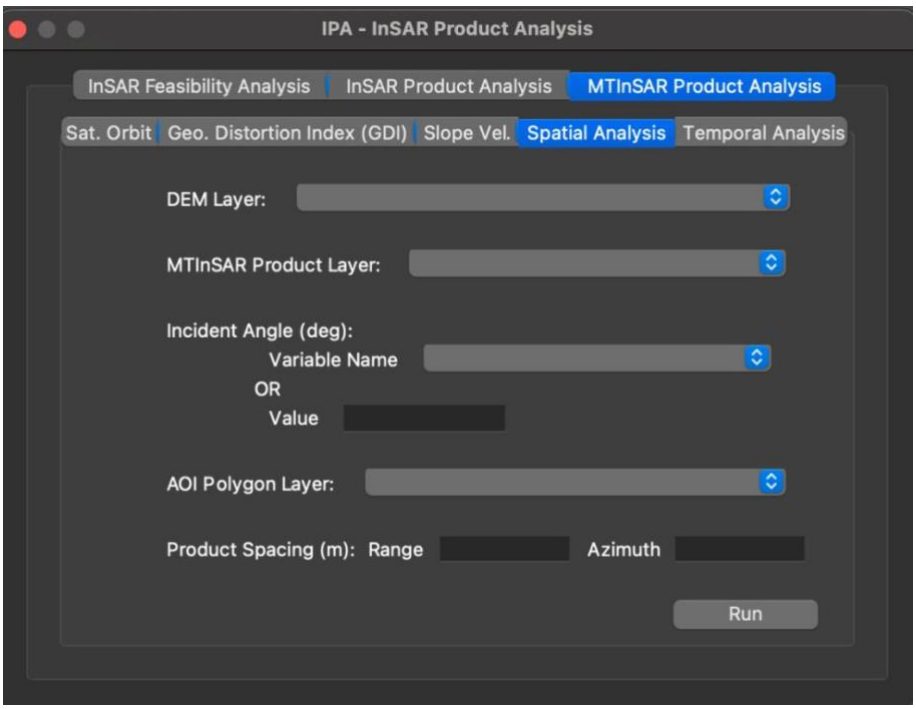


Figure 11 *– MTInSAR Product Analysis window – Spatial Analysis tab*

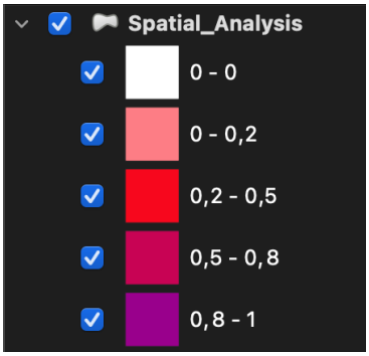


Figure 12 *– Default style applied to a Spatial Analysis output*

5.3.5 Temporal Analysis

This functionality performs a displacement time series analysis based on automated procedures recently developed for identifying nonlinear signals and based on Fuzzy Entropy (FE) and Fisher statistics (Bovenga et al., 2021). This allows identifying coherent targets affected by nonlinear displacements, such as the accelerating movements of the pre-failure phase of a landslide.

The following inputs are required:

- ***MTInSAR Product Layer*****:** vector layer containing the MTInSAR products (e.g. coherent target locations, mean rate LOS displacement values in millimetres per year, LOS displacement time series in millimetres);
- ***Displacement Prefix*****:** prefix of the MTInSAR layer that indicates the displacement fields;
- ***Date Format*****:** format for the date used in the displacement field names and selected from a preconfigured dropdown menu (Table 2);
- ***Wavelength***: wavelength of the SAR signal, expressed in millimeters;
- ***LOS Vel. Name***: name of the field corresponding to the LOS velocity in millimetres per year and selected from the list of MTInSAR layer fields;

- ***ID*:** field name of the MTInSAR layer that uniquely identifies each coherent target;
- ***Coherence*:** name of the field of the MTInSAR layer that corresponds to the temporal coherence value associated to each target;

Date format presets
dd/MM/yy
MM/dd/yy
yyyy-MM-dd
ddMMyyyy
yyyyMMdd

Table 2 *– Date format presets available in the plugin*

Output:

The function returns a vector layer, a copy of the input MTInSAR layer, which also includes the following attributes for each coherent target:

Field	Description
FE1 – FE5	Fuzzy Entropy coefficients (dimensions 1 to 5), computed on the displacement time series to assess its degree of regularity and local predictability.
PolSel	Integer flag (1–5) identifying the optimal kinematic regime assigned to the target by the hierarchical Model Selection algorithm.
c0	Deterministic intercept: estimated displacement at the reference epoch (t_0).
c1	First-degree coefficient, associated with the stationary linear deformation rate of the point.
c2	Second-degree coefficient, linked to the acceleration/deceleration kinematic factor of the landslide or glacial body.
c3	Third-degree coefficient, modeling temporary variations in displacement rate and impulsive accelerations.
c4	Fourth-degree coefficient, reproducing complex, non-linear deformation components.
RMSE	Root Mean Square Error (in mm) of the residuals of the selected polynomial fit.
R2	Coefficient of Determination ($R^2 \in [0,1]$), the share of displacement variance explained by the regression.
F	Global regression quality indicator, based on the Fisher F-statistic distribution.
FA	F-change test statistic, used to verify whether increasing the polynomial order is justified by a real increase in explained variance.
AIC	Corrected Akaike Information Criterion, used to penalize model over-parameterization (overfitting).
Coh	Residual temporal coherence, mapping the regression error metric back into the native domain of the SAR interferometric phase, to verify the model's physical accuracy.

Table 3 *– Output attribute fields produced by Temporal Analysis*

The PolSel field classifies each target according to the following kinematic regimes:

- 1: linear trend at constant velocity;
- 2: quadratic trend, with uniform acceleration or deceleration;
- 3: cubic trend, with an inflection point or an abrupt reversal of motion;
- 4: quartic trend, with complex cyclical oscillations;
- 5: unclassified target, dominated by stochastic error.

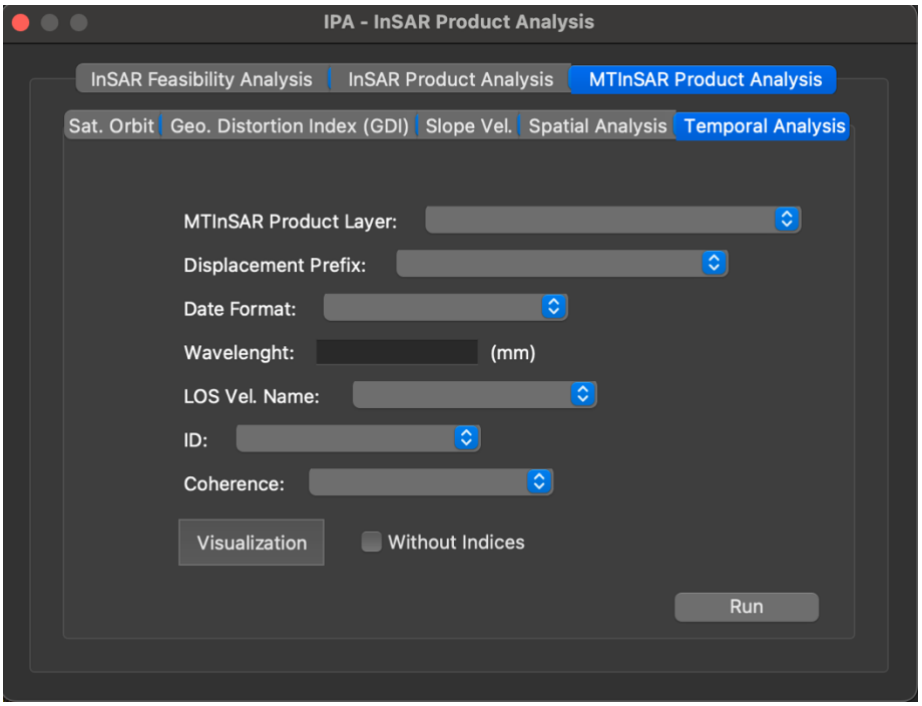


Figure 13 *– MTInSAR Product Analysis window – Temporal Analysis tab*

Visualization

By clicking the Visualization button, the user activates an interactive map inspection tool to explore displacement time series directly on the canvas. The inspection mode is controlled via the *Without Indices* checkbox:

- **Unchecked (Default / With Indices)**: plots the time series of the selected coherent target and displays all calculated statistical indices (e.g. polynomial coefficients, RMSE, R^2) alongside the chart;
- **Without indices**: plots the time series showing only basic information — coherence, velocity, and ID of the selected PS.

Once the tool is active, the operator can click directly on a target on the QGIS map canvas: the plugin instantly extracts the historical displacement values of that point and generates a high-resolution chart showing the measured time series, the fitted polynomial trend curve selected through the statistical indices, and, if requested, the corresponding accuracy and error indices.